

Artificial Intelligence

AI2002

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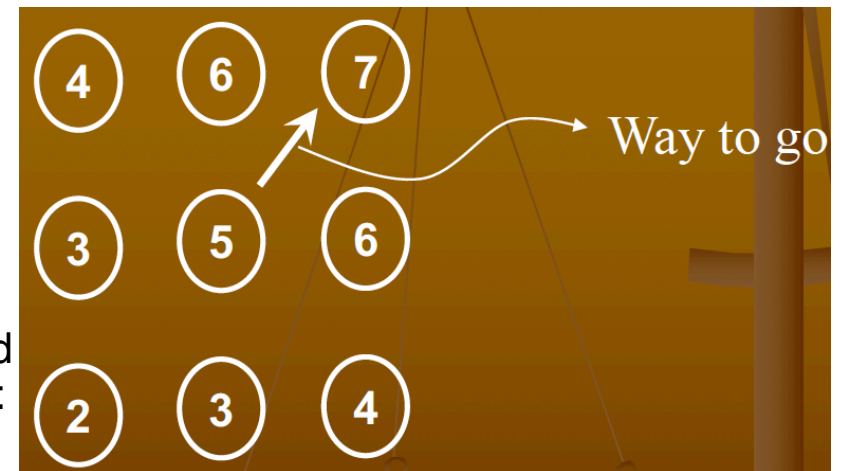
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Local Search: Hill Climbing

- The simplest way to implement heuristic search is through a procedure called hill-climbing
- Hill-climbing strategies expand the current state of the search and evaluate its children.
- The “best” child is selected for further expansion; neither its siblings nor its parent are retained.
- Hill climbing is named for the strategy that might be used by an eager, but blind mountain climber: go uphill along the steepest possible path until you can go no farther up.
- Because it keeps no history, the algorithm cannot recover from failures of its strategy.

Hill climbing search (Real-world Analogy)

- In the case where only local knowledge is available - you have no idea how far away the goal is - the hill-climbing method may be a good solution.
- The method is best described as if you are trying to climb a hill in fog.
- You have an altimeter in your pocket and you want to go up.
- The way the **hill-climbing heuristic** works is to **find the altitude of places near your current location** and then **going in the direction** that gives you the **highest upward gradient**.
- The term local knowledge describes the fact that you know the altitude of places in your locality, the local terrain.



The numbers in circles indicate the altitudes of neighboring places.

Example: tic tac toe

The algorithm only picks the state with max possible wins at each level

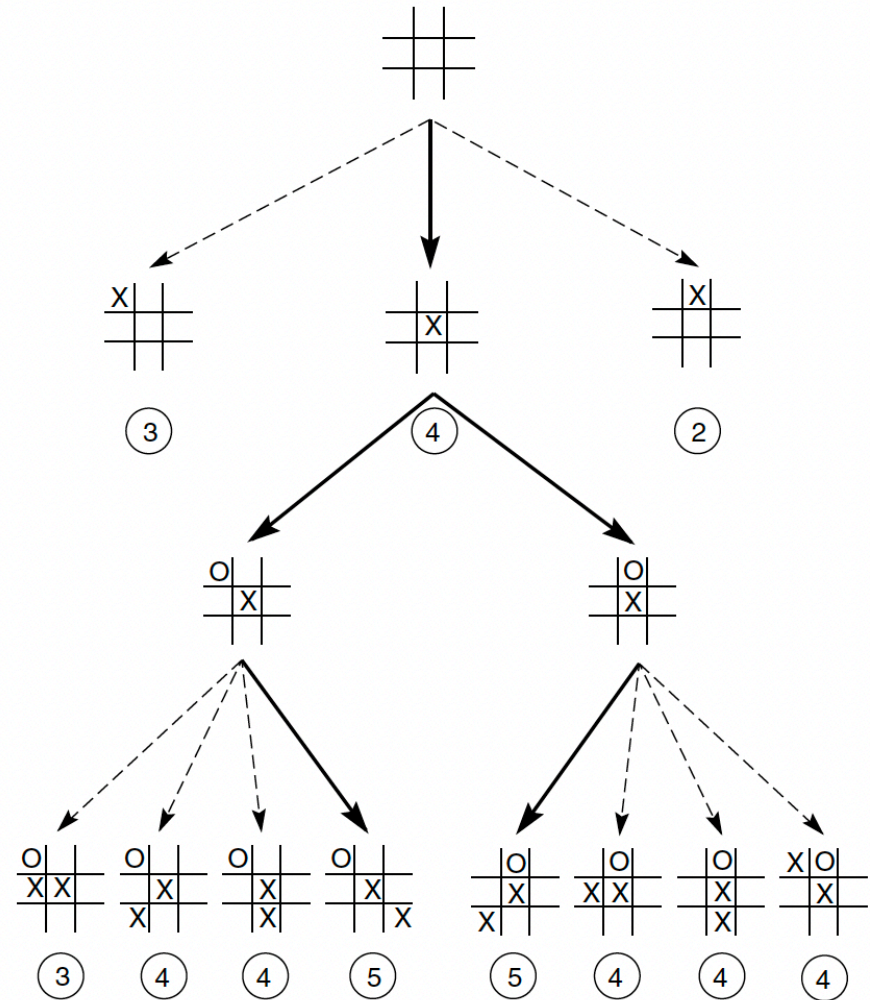


Figure 4.3 Heuristically reduced state space for tic-tac-toe.

Local Search: Hill Climbing

- A major problem of hill-climbing strategies is their tendency to become stuck at local maxima.
- If they reach a state that has a better evaluation than any of its children, the algorithm falters.
- If this state is not a goal, but just a local maximum, the algorithm may fail to find the best solution.
- That is, performance might well improve in a limited setting, but because of the shape of the entire space, it may never reach the overall best.

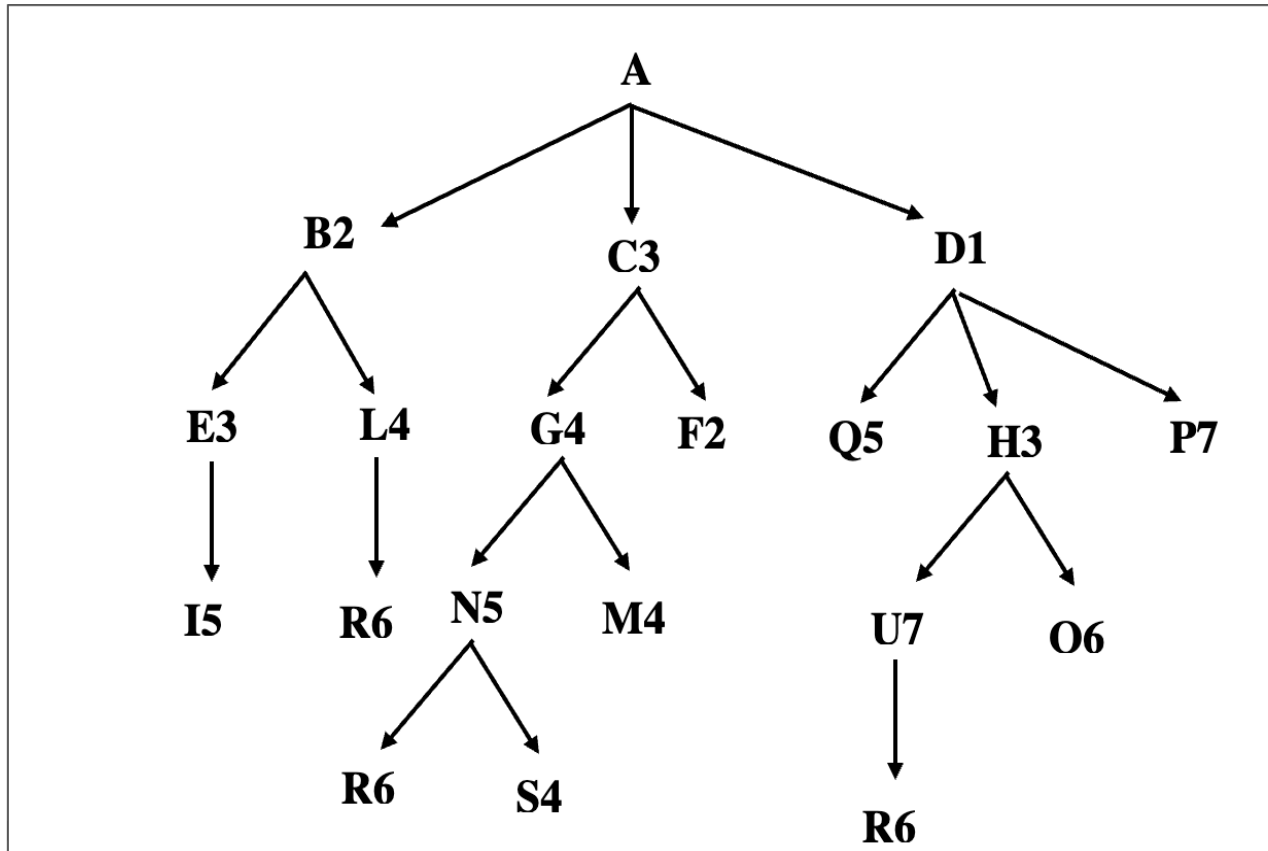
Example

- An example of local maxima in games occurs in the 8-puzzle.
- Often, in order to move a particular tile to its destination, other tiles already in goal position need be moved out.
- This is necessary to solve the puzzle but temporarily worsens the board state.
- Because “better” need not be “best” in an absolute sense, search methods without backtracking or some other recovery mechanism are unable to distinguish between local and global maxima.

Best-First vs. Hill Climbing

- This procedure looks very similar to hill climbing, with two exceptions:
 1. In **hill climbing**, **one move is selected** and all other moves are **rejected**, never to be considered.
- In best-first search, **one move is selected**, but the others are kept around so that they can be **revisited** later if the selected path becomes **less promising**.
 2. The **best available state** is selected in best-first search, even if that state has a value that is lower than the value of the state that was just explored.
- This is in contrast with hill climbing, which will stop if there are no successor states with better values than the current state.

Example: A to R (maximization)



Solution

Hill climbing doesn't store the siblings or parents nodes for further exploration, it picks the best option from the local search

[OPEN]	[CLOSED]	X 
[A]	[]	A
[C3,B2,D1]	[A]	C3
[G4,F2]	[C3,A]	G4
[N5,M4]	[G4,C3,A]	N5
[R6,S4]	[N5,G4,C3,A]	R6
[]	[R6, N5,G4,C3,A]	

Final path: A → C3 → G4 → N5 → R6